

Question 1 [int8 matmul]: Say we want to do the matmul

$X[B,D] \cdot DY[D,F] \rightarrow Z[B,F]$ in int8 precision (1 byte per parameter) instead of bfloat16 (2 bytes per parameter),¹⁴ since TPUs/GPUs can do matmuls faster in lower precision.

1. How many bytes need to be loaded from memory? How many need to be written back to memory?
2. How many total OPs are performed?
3. What is the arithmetic intensity?
4. What is a roofline estimate for T_{math} and T_{comms}
5. What are reasonable upper and lower bounds for the runtime of the whole operation?

Assume our HBM bandwidth is $8.1e11$ bytes/s and our int8 peak OPs/s is $3.94e14$ (about 2x bfloat16).

My Answer —

- 1) $BD+DF$ loaded and BF write back
- 2) nearly $2*BDF$ in int8 peak OP
- 3) $2*BDF/(BD+DF+BF)$ which can be approximated to $2*B$ if $B \ll D$ and $B \ll F \rightarrow$ if $B > (3.94e14)/(2*8.1e11)$ $B > 243$
- 4) $T_{\text{math}} \sim 2*BDF/3.94e14$ $T_{\text{comms}} \sim (BD+DF+BF)/8.1e11$
- 5) upper bound is $T_{\text{math}}+T_{\text{comms}}$ and lower bound is $\max(T_{\text{math}}, T_{\text{comms}})$

Question 2 [int8 + bf16 matmul]: In practice we often do different weight vs. activation quantization, so we might store our weights in very low precision but keep activations (and compute) in a higher precision. Say we want to quantize our weights in int8 but keep activations (and compute) in bfloat16. At what batch size do we become compute bound? Assume $1.97e14$ bfloat16 FLOPs/s.

Hint: this means specifically $\text{bfloat16}[B, D] * \text{int8}[D, F] \rightarrow \text{bfloat16}[B, F]$ where B is the "batch size".

My Answer —

$AI(\text{arithmetic intensity}) = \text{num flops} / \text{num bytes comm}$

AI for TPU=240

$\text{numflops} = 2*BDF$ (because int will also be treated as float)

num bytes comm= $2*BD+DF+2*BF \approx DF$
 $AI=2*B=240 \Rightarrow B>120$

Question 3: Taking the setup from Question 2, make a roofline plot of peak FLOPs/s vs. B for $F=D=4096$ and $F=D=1024$. Use the exact number of bytes loaded, not an approximation

My Answer 

<https://www.desmos.com/calculator/du3bccatee>

`<iframe src="https://www.desmos.com/calculator/hrgrfz5hkt?embed" width="500" height="500" style="border: 1px solid #ccc" frameborder=0></iframe>`

use above embed for visualisation

Question 4: What if we wanted to perform $\text{int8}[B, D] * \text{Dint8}[B, D, F] \rightarrow \text{int8}[B, F]$ where we imagine having a different matrix for each batch element. What is the arithmetic intensity of this operation?

My answer:-

Bytes:- $BD+BDF+BF$

ops:- $2*BDF$ (ints ops)

$AI = 2*DF/(D+F+DF)$ $DF \gg D, F \Rightarrow AI=2$

Problem 5 [Memory Rooflines for GPUs]: Using the [spec sheet provided by NVIDIA for the H100 SXM](#), calculate the batch size at which a bfloat16 matrix multiplication will become compute-bound. Note that the Tensor Core FLOPs numbers are twice the true value since they're only achievable with structured sparsity.

BFLOAT16 Tensor Core*	1,979 teraFLOPS
GPU Memory Bandwidth	3.35TB/s

$B > \text{BFLOAT16}/(2*BW) \Rightarrow B > 295$